

A Two-Stage Perceptual Account of Interval Consonance

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Abstract

Euler’s *Gradus Suavitatis* (1739) assigns a dissonance value to a musical interval p/q by the formula $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, where $\Omega^*(n) = \sum_i e_i(p_i - 1)$ sums the weighted prime exponents of n . We show that the simpler asymmetric formula $f(p/q) = p + \Omega^*(q)$ achieves a higher rank correlation with human consonance ratings ($\rho = 0.989$ vs. 0.979), and that this formula admits a natural two-stage perceptual interpretation grounded in harmonic series theory. We also show that *Gradus* is exactly equivalent to a weighted harmonic coincidence count with weights $w(n) = \Omega^*(n)$, and that the simpler metric $\max(p, q)$ — equivalent to Galileo’s pulse-coincidence model — achieves the same correlation as $f(p/q)$.

1 Background

Euler’s *Gradus Suavitatis*, catalogued in OEIS A275314, assigns to each musical interval a positive integer measuring its arithmetic dissonance. For a ratio p/q in lowest terms, the characteristic number is $n = p \cdot q$, and:

$$G(p/q) = 1 + \Omega^*(p \cdot q)$$

where Ω^* is the completely additive function $\Omega^*(n) = \sum_i e_i(p_i - 1)$. Since Ω^* is completely additive, $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, treating numerator and denominator symmetrically. The weights $(p_i - 1)$ for prime p_i encode Euler’s aesthetic judgment that large primes represent greater “difficulty.”

Euler’s formula correctly ranks the 13 standard intervals of Western music against human consonance ratings [7] with Spearman $\rho = 0.979$, and is exactly tied only on the major third / major sixth pair and on a three-way tie at *Gradus* 8.

2 Gradus as Harmonic Coincidence

For two notes at frequency ratio $p : q$ (coprime), the m -th harmonic coincidence occurs at absolute frequency $mp \cdot f_0$, which is the (mp) -th harmonic of the lower note and the (mq) -th harmonic of the upper note. Assigning weight $w(n) = \Omega^*(n)$ to the n -th harmonic of each note, the weighted contributions to the lower and upper note respectively, summed over M coincidences, are:

$$\begin{aligned} \text{Lower: } & \sum_{m=1}^M \Omega^*(mp) = \Omega^*(p) \cdot M + C(M) \\ \text{Upper: } & \sum_{m=1}^M \Omega^*(mq) = \Omega^*(q) \cdot M + C(M) \end{aligned}$$

where $C(M) = \sum_{m=1}^M \Omega^*(m)$ is independent of both p and q . Both identities follow from the complete additivity of Ω^* : $\Omega^*(mn) = \Omega^*(m) + \Omega^*(n)$ for all positive integers m, n . Adding the

two sums gives the symmetric score:

$$\text{Score} = M \cdot (\Omega^*(p) + \Omega^*(q)) + 2C(M) = M \cdot (G(p/q) - 1) + 2C(M)$$

Gradus is therefore exactly a weighted harmonic coincidence count, with harmonics weighted by their own prime complexity. This confirms the physical intuition behind Euler’s formula but reveals its self-referential structure: recovering Gradus from coincidences requires weights that are themselves Gradus-like.

Unweighted coincidence counting (flat weights) achieves $\rho = 0.791$ across all coprime ratios with $p, q \leq 16$. The weight function $\log n$ achieves $\rho = 0.795$ without any prime structure, establishing a ceiling for physics-based models that do not invoke prime factorisation.

3 Galileo’s Model and $\max(p, q)$

Galileo (1638) described consonance in terms of pulse coincidences: two strings at ratio $p : q$ produce coinciding pulses every p vibrations of the lower string. The “sweetness” of the interval grows with the frequency of coincidences, i.e. inversely with p . The metric $\max(p, q) = p$ is therefore:

1. **Galileo’s pulse model** — vibrations between coincidences;
2. **Harmonic series position** — the upper note is partial p of the implied fundamental [4];
3. **Farey index** — the smallest n such that q/p appears in the Farey sequence F_n (Farey 1816; proved by Cauchy);
4. **First coincidence harmonic** — the position in the lower note’s series where the two notes first meet.

These four descriptions, discovered independently across four centuries, are mathematically identical. $\max(p, q)$ achieves $\rho = 0.989$ against human ratings, marginally exceeding Gradus, but cannot distinguish intervals with the same numerator (e.g. $5/4$ and $5/3$ both have $\max = 5$).

4 An Improved Formula

Throughout this paper, p/q denotes a frequency ratio in lowest terms with $p > q \geq 1$ and $\gcd(p, q) = 1$, so that $p/q > 1$ represents an interval above unison and p is always the numerator (upper note).

Euler’s formula is symmetric: $\Omega^*(p) + \Omega^*(q)$. Human ratings, however, distinguish numerator and denominator — the bass note and the upper note play different perceptual roles. We find empirically that the asymmetric formula:

$$\boxed{f(p/q) = p + \Omega^*(q)} \quad (p > q \geq 1, \gcd(p, q) = 1)$$

achieves $\rho = 0.989$, matching $\max(p, q)$ and exceeding Gradus. The formula differs from Gradus only in replacing $\Omega^*(p)$ with p for the numerator. Optimising prime weights freely with six parameters — separate weights $w_p(r)$ and $w_q(r)$ for each prime $r \in \{2, 3, 5\}$ in numerator and denominator respectively — achieves perfect rank correlation $\rho = 1.000$. The fitted weights are:

$$\begin{aligned} w_p(2) &= 2.64, & w_p(3) &= 4.44, & w_p(5) &= 8.62 \\ w_q(2) &= 2.40, & w_q(3) &= 6.38, & w_q(5) &= 7.70 \end{aligned}$$

The denominator weights for primes 3 and 5 are substantially larger than their numerator counterparts ($w_q(3)/w_p(3) \approx 1.44$, $w_q(5)/w_p(5) \approx 0.89$), consistent with the view that the denominator requires stronger prime-sensitivity than the numerator. The formula f , which sets $w_p(r) = r - 1$ and $w_q(r) = r - 1$ for the denominator while replacing the numerator sum with the raw integer p , is a principled simplification of this six-parameter family.

The formula $f(p/q) = p + \Omega^*(q)$ decomposes as the sum of two quantities:

Component	Formula	Meaning
Upper note reach	p	Partial number of upper note in implied harmonic series
Bass context depth	$\Omega^*(q)$	Prime complexity of lower note's position in series

5 A Two-Stage Perceptual Model

The formula motivates a two-stage model of how the auditory system evaluates an interval:

Stage 1 — Bass establishes harmonic context (cost $\Omega^*(q)$). The lower note is the q -th partial of an implied fundamental F . The brain extrapolates downward to F by “dividing out” the prime factors of q . Each factor of 2 (an octave) costs 1 unit; each factor of 3 costs 2; each factor of 5 costs 4. This is exactly $\Omega^*(q)$. Intervals whose bass note is a power of 2 (e.g. the octave, $q = 1$; the major second bass, $q = 8 = 2^3$) establish the fundamental cheaply via octave equivalence. Intervals with an odd prime in q (the fourth, $q = 3$; the minor third bass, $q = 5$) require the brain to extrapolate through a genuinely new prime, incurring higher cost.

Stage 2 — Upper note reaches into the series (cost p). Once the fundamental is established, the upper note must be recognised as its p -th partial. Higher partials are quieter (amplitude $\propto 1/p$ for harmonic sounds with a falling spectral envelope, such as bowed strings and the singing voice; the exact roll-off is instrument-dependent), more narrowly tuned, more sensitive to mistuning, and have lower pitch salience [6]. The cost of confirming the upper note as a member of the harmonic series is therefore p itself.

Total perceptual effort: $f(p/q) = p + \Omega^*(q)$.

This account connects:

- **Galileo** (1638): pulse coincidences as physical mechanism;
- **Rameau** (1722): the *corps sonore* (harmonic series) as the source of musical meaning;
- **Euler** (1739): prime arithmetic as the language of harmonic complexity;
- **Terhardt** (1979): virtual pitch and harmonic template matching as auditory mechanisms;
- **Huron** (2001): statistical learning of harmonic patterns as the basis of consonance judgments.

6 Limitations: the Major Third and Major Sixth Tie

No simple formula in this class distinguishes the major third ($5/4$, human rank 5) from the major sixth ($5/3$, human rank 7). Both have $p = 5$ and $\Omega^*(q) = 2$ (since $\Omega^*(4) = \Omega^*(2^2) = 2$ and $\Omega^*(3) = 2$). The distinction requires knowing that $q = 4 = 2^2$ is simpler than $q = 3$ under octave equivalence, yet the prime-weighting function cannot see this.

The Tenney height $H(p/q) = p \cdot q$ [11] — the standard complexity measure in just intonation theory — makes the situation worse: $H(5/4) = 20 > H(5/3) = 15$, so Tenney ranks the major sixth as *more consonant* than the major third, the opposite of human judgments. Gradus, f ,

and max all tie the pair; Tenney actively inverts them. This failure of Tenney height is a direct consequence of its symmetric and multiplicative structure: $H(p/q) = p \cdot q$ is dominated by whichever of p and q is larger, which for $5/3$ is $q = 3$ (small) rather than for $5/4$ where $q = 4$ (larger). The asymmetric formula f ties them for the same structural reason — $\Omega^*(3) = \Omega^*(4) = 2$ — but at least does not invert the order.

Lahdelma & Eerola [8] decompose consonance ratings into sensory and familiarity components; on *sensory consonance* alone, $5/4$ and $5/3$ are nearly indistinguishable across listeners, suggesting that the tie in $f(p/q)$ may accurately reflect the purely arithmetic component of the percept. The gap observed in Krumhansl’s data is plausibly attributable to learned exposure — major thirds occur far more frequently than major sixths in tonal music — consistent with the finding of McDermott et al. [9] that consonance preferences in a culture with minimal Western music exposure are substantially reduced. The converse view — that consonance preferences reflect universal statistics of harmonic sounds in the environment, including voiced speech — is advanced by Bowling, Purves & Gill [12], who find that vocal similarity predicts interval attraction across cultures; on this account the residual consonance of ET thirds may reflect their partial alignment with vocal harmonic spectra. Resolving the tie within an arithmetic formula would therefore require modelling cultural familiarity, which is outside the scope of the present approach.

7 Summary

Formula	ρ vs. human	Notes
Tenney height: $H = p \cdot q$	0.978	Symmetric; inverts $5/4$ vs $5/3$
Euler’s Gradus: $1 + \Omega^*(p) + \Omega^*(q)$	0.979	Symmetric; two tie groups
Galileo / $\max(p, q)$	0.989	Simplest; one tie
$f(p/q) = p + \Omega^*(q)$	0.989	Asymmetric; two-stage interpretation
Optimised asymmetric weights	1.000	Six parameters; no closed form
Roughness (Plomp–Levelt)	0.709–0.863	Register-dependent; fails on tritone*

* Helmholtz [3] introduced the beating-based dissonance curve that Plomp & Levelt [5] later parametrised from listener data; the two models produce equivalent rankings for our purposes.

Table 1 shows all three formulae alongside the human consonance rank for the 13 standard just intervals of Western music, ordered by human rating. Ties in $f(p/q)$ and in $\max(p, q)$ are indicated by matching values; the sole unresolved tie (major third / major sixth, both $f = 7$) is discussed in Section 6.

8 Comparison of the Three Formulae

Table 1 invites a direct comparison of the three arithmetic formulae. Several observations follow.

Agreement on the extremes. All three formulae agree without qualification on the most and least consonant intervals. The unison (1), octave (2), fifth ($3/2$), and fourth ($4/3$) are assigned the lowest values by every measure, consistent with their historical priority in counterpoint and their prominence in non-Western musical systems. The tritone ($45/32$) receives the highest value under every formula, and the minor and major seconds cluster near the top of the dissonance scale in all three cases. The disagreements are concentrated in the middle of the table, among intervals that are also the most contested in historical music theory.

Table 1: Dissonance values for the 13 standard just intervals, ordered by human consonance rank [7] (rank 1 = most consonant).

Interval	Ratio	$G(p/q)$	$f(p/q)$	$\max(p, q)$	$H(p/q)$	Human rank
Unison	1/1	1	1	1	1	1
Octave	2/1	2	2	2	2	2
Fifth	3/2	4	4	3	6	3
Fourth	4/3	5	6	4	12	4
Major third	5/4	7	7	5	20	5
Minor third	6/5	8	10	6	30	6
Major sixth	5/3	7	7	5	15	7
Minor sixth	8/5	8	12	8	40	8
Major second	9/8	8	12	9	72	9
Minor seventh	9/5	9	13	9	45	10
Major seventh	15/8	10	18	15	120	11
Minor second	16/15	11	22	16	240	12
Tritone	45/32	14	50	45	1440	13

$H(p/q) = p \cdot q$ is the Tenney height [11], the standard complexity measure in just intonation theory.
Spearman ρ against human rank: $G = 0.979$, $f = 0.989$, $\max = 0.989$, $H = 0.978$.

Correlation and ties. Euler’s Gradus achieves a Spearman rank correlation of $\rho = 0.979$ against the human data; f and $\max$ both achieve $\rho = 0.989$. The difference is modest, and caution is warranted given the small sample (13 intervals) and the cultural specificity of the rating data (see Section 6). More informative than the correlation coefficient is the tie structure. Gradus produces two tie groups among the 13 intervals: the major third and major sixth share $G = 7$, and the minor third, minor sixth, and major second share $G = 8$. The three-way tie at $G = 8$ is particularly awkward, as the three intervals span three consecutive human ranks (6, 8, 9). The formula f resolves this three-way tie — giving $f = 10, 12, 12$ respectively — though it introduces a two-way tie between the minor sixth and major second (both $f = 12$, human ranks 8 and 9). The metric $\max(p, q)$ shares the major third / major sixth tie with f , but adds a further tie between the major second and minor seventh (both $\max = 9$, human ranks 9 and 10). On the criterion of tie-resolution among these 13 intervals, f performs best, Gradus intermediate, and $\max$ least well.

Dynamic range and the tritone. The three formulae differ most sharply in their treatment of the tritone. Gradus gives $G(45/32) = 14$, only four units above the major seventh ($G = 10$). Both f and $\max$ assign substantially higher values ($f = 50$, $\max = 45$), placing the tritone in a category of its own and reflecting the large numerator $p = 45$ that the asymmetric formulae weight directly. Whether this wider dynamic range better represents human experience is difficult to assess from rank-correlation data alone; what can be said is that the tritone is consistently rated as by far the most dissonant interval in the human data, and both f and $\max$ capture that separation more emphatically than Gradus.

A caveat is necessary: the tritone used throughout this paper is $45/32$, the 5-limit augmented fourth generated within the diatonic scale as $45/32 = (9/8) \cdot (5/4)$. This choice is conventional in 5-limit just intonation but is not the only defensible one. The septimal tritone $7/5$ (582.5 cents) gives $f(7/5) = 7 + \Omega^*(5) = 11$ and $G(7/5) = 11$ — barely more dissonant than the major third under either measure. The undecimal tritone $11/8$ gives $f = 14$. These alternatives sit in different prime limits and carry different musical connotations: $7/5$ is the tritone of barbershop and some jazz practice; $45/32$ is the tritone of classical diatonic harmony. The extreme values $f = 50$ and $\max = 45$ are therefore properties of the 5-limit diatonic tritone specifically, not of the

tritone concept in general, and should be interpreted accordingly. The high values are, however, consistent with the 5-limit diatonic context in which the human rating data were collected.

Symmetry. Gradus is symmetric in p and q : $G(p/q) = G(q/p)$. This reflects Euler’s focus on the product $p \cdot q$ as the relevant arithmetic quantity, and is appropriate if the perceptual roles of the two notes are interchangeable. Both f and $\max$ are asymmetric, treating the numerator (upper note) and denominator (lower note) differently, as is the case when one note functions as a bass and the other as a melody note. The symmetry of Gradus means it cannot distinguish $9/5$ from $16/9$ — two minor sevenths that differ by only 1 cent but arise from different tuning systems — whereas f gives $f(9/5) = 13$ and $f(16/9) = 20$. Whether this asymmetry reflects a genuine perceptual distinction or an artefact of the formula is an open question, though the perceptual account in Section 5 offers a principled motivation.

Interval inversion. Two intervals whose frequency ratios multiply to 2 are *inversions* of each other within the octave: the major third ($5/4$) inverts to the minor sixth ($8/5$), the fifth ($3/2$) to the fourth ($4/3$), and so on. Under Gradus and under Tenney height, inversionally related intervals receive similar but not identical values ($G(5/4) = 7$ vs $G(8/5) = 8$; $H(5/4) = 20$ vs $H(8/5) = 40$). The formula f is more strongly asymmetric: $f(5/4) = 7$ but $f(8/5) = 12$, a gap of 5 units. This captures the perceptual asymmetry — a minor sixth in root position (bass on the lower note) is generally heard as less stable than a major third — but amplifies it considerably. Whether the magnitude of the asymmetry is empirically justified would require testing beyond the Krumhansl dataset, which does not distinguish root-position from inverted voicings.

Tenney height. A fourth measure absent from the summary table but standard in just intonation theory is the Tenney height $H(p/q) = p \cdot q$ [11], also known as the Benedetti height. It achieves $\rho = 0.978$ against the human data, comparable to Gradus and slightly below f and $\max$. Tenney height is symmetric like Gradus, but multiplicative rather than additive: it measures the prime complexity of the product $p \cdot q$ without distinguishing numerator from denominator. Its ordering of the 13 intervals differs from human judgments most sharply on the major third / major sixth pair: $H(5/4) = 20 > H(5/3) = 15$, placing the major sixth above the major third in consonance — directly opposite to human ratings. This failure is qualitatively different from the tie produced by f and $\max$: all additive prime-weighted formulae tie $5/4$ and $5/3$ because $\Omega^*(4) = \Omega^*(3)$, whereas Tenney actively inverts them. The result supports the view that a multiplicative symmetric measure is not the right structure for this problem.

Simplicity and interpretability. The metric $\max(p, q)$ is strikingly simple, requires no prime factorisation, and admits four independent mathematical characterisations (Section 3). These properties make it an attractive baseline. Its weakness is precisely that simplicity: it is insensitive to whether the denominator is a power of 2 or a large prime, treating $4/3$ and $7/3$ identically in the denominator. Gradus captures this prime sensitivity through $\Omega^*(q)$ but applies the same weighting to the numerator, where the evidence suggests a different treatment is warranted. The formula f retains prime sensitivity in the denominator while replacing the numerator term with the raw partial number, motivated by the two-stage perceptual account. Each formula thus represents a different theory of what makes an interval consonant, and the empirical data do not decisively separate them.

Sensitivity near just ratios. Figure 1 illustrates these differences directly. For two showcase intervals — the fifth ($3/2$) and the major third ($5/4$) — every rational number with denominator ≤ 32 within ± 40 cents of the just ratio is plotted against its formula value. The just ratio sits at the centre of a deep trough, surrounded by rationals with substantially higher values under all

three formulae. The red markers (f) are consistently the tallest: moving roughly 20 cents from the just fifth, f reaches values of 70 or above while Gradus (G , blue) remains in the 40–55 range. The green markers (max) are intermediate but track the numerator p directly and so cluster at values that increase smoothly with denominator size. Within the ± 15 cent tolerance zone (gold shading), no rival rational with denominator ≤ 32 exists for either interval, confirming that the just ratios are genuinely isolated at low complexity in their neighbourhood.

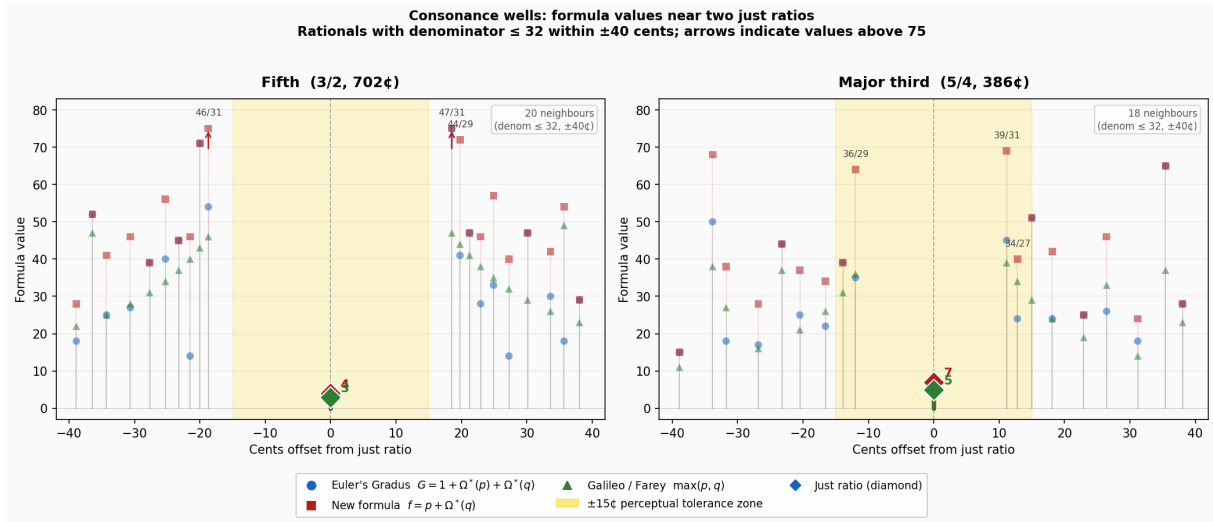


Figure 1: Formula values for all rationals with denominator ≤ 32 within ± 40 cents of the just fifth ($3/2$, left) and major third ($5/4$, right). Diamond markers show the just ratio; circles, squares, and triangles show neighbouring rationals under G , f , and $\max$ respectively. Upward arrows indicate values above 75. The gold band marks the ± 15 cent perceptual tolerance zone; no rival rational falls within it for either interval.

Choice of formula. The three formulae are more complementary than competing. For a context requiring only a rough ordering of standard intervals, any of the three suffices. Where fine distinctions matter — competing tuning systems, intervals outside the standard 13, or analysis of the consonance landscape under equal temperament — the formulae diverge in informative ways, and the choice should be guided by the theoretical question at hand.

9 The Consonance Landscape and Equal Temperament

The three formulae are defined on rational numbers, but musical intervals in practice are never exact. To study how each formula behaves near a just ratio we sweep the octave in 1-cent steps, find the simplest rational approximation p/q (denominator ≤ 64) within ± 15 cents of each position, and evaluate G , f , and $\max$. The result is a *consonance landscape* — a spiky function of pitch interval in which low values cluster sharply around simple fractions and rise steeply on either side.

Well depth and sharpness

Table 2 shows, for each standard just interval, the nearest rival rational within ± 50 cents and the corresponding jump in each formula. The wells around simple fractions are deep for all three functions, but $f(p/q)$ is substantially steeper: moving 2–5 cents off the fifth ($3/2$) takes f from 4 to 104 (+100), compared with Gradus rising from 4 to 58 (+54). The major third is even more extreme: 3 cents off $5/4$, f leaps from 7 to 136, while Gradus rises from 7 to 81. This

sharper gradient means $f(p/q)$ predicts tighter perceptual boundaries around just intonation than Gradus does — a potentially testable difference.

Table 2: Nearest rival rational for each just interval and the resulting jump in each formula. “Rival” is the simplest fraction within ± 50 cents that is distinct from the just ratio; error is the rival’s distance in cents.

Interval	Just	Rival	Err (¢)	ΔG	Δf	$\Delta \max$
Major second	9/8	64/57	2	+19	+72	+55
Minor third	6/5	77/64	2	+15	+73	+71
Major third	5/4	76/61	3	+74	+129	+71
Fourth	4/3	83/62	4	+109	+108	+79
Tritone	45/32	52/37	1	+37	+38	+7
Fifth	3/2	94/63	5	+54	+100	+91
Minor sixth	8/5	101/63	2	+103	+99	+93
Major sixth	5/3	103/62	3	+127	+127	+98
Minor seventh	9/5	16/9	1	0	+7	+7
Major seventh	15/8	118/63	1	+60	+110	+103

Equal temperament

Twelve-tone equal temperament (12-TET) divides the octave into 12 equal semitones of 100 cents each; no interval except the unison and octave is exactly just. Table 3 gives the nearest just approximation for each 12-TET semitone and the formula values at that approximation.

Table 3: 12-TET semitones and their 5-limit just intonation targets. Error = ET cents – just cents (positive = ET is sharp of just). Formula values are evaluated at the just target, not the ET pitch.

Note	ET (¢)	Just target	Just (¢)	Error (¢)	G	f	H
C (unison)	0	1/1	0.0	0.0	1	1	1
C#	100	16/15	111.7	−11.7	11	22	240
D	200	9/8	203.9	−3.9	8	12	72
E♭	300	6/5	315.6	−15.6	8	10	30
E	400	5/4	386.3	+13.7	7	7	20
F	500	4/3	498.0	+2.0	5	6	12
F#	600	45/32	590.2	+9.8	14	50	1440
G	700	3/2	702.0	−2.0	4	4	6
A♭	800	8/5	813.7	−13.7	8	12	40
A	900	5/3	884.4	+15.6	7	7	15
B♭	1000	9/5	1017.6	−17.6	9	13	45
B	1100	15/8	1088.3	+11.7	10	18	120
C (octave)	1200	2/1	1200.0	0.0	2	2	2

Each 12-TET semitone is shown alongside its standard 5-limit just intonation target — the simple ratio that the tempered interval approximates in diatonic tonal music. The formula values (evaluated at the just target) reveal a clear division: the fourth and fifth have small errors (2 cents) and low formula values; the thirds and sixths have errors of 14–16 cents and moderate values; the minor second, minor seventh, and tritone have both large errors and the highest complexity values. The equal-tempered major third (400 cents) is 14 cents sharp of its just target 5/4 ($f = 7$); the equal-tempered minor seventh (1000 cents) is 18 cents flat of 9/5 ($f = 13$).

In each case, the ET semitone relies on the ear pulling a mistuned pitch into the low-complexity well of its just target — the same well structure examined in Section 8. Note that the F# entry uses 45/32, the 5-limit diatonic tritone; as discussed in Section 8, the septimal tritone 7/5 ($f = 11$) would be a musically different just target.

Alternative equal temperaments

The tolerance model makes a natural prediction about which equal divisions of the octave (EDOs) are acoustically viable for which intervals. An n -EDO divides the octave into n equal steps of $1200/n$ cents; an interval is *well-approximated* when the nearest step falls within Δ_{tol} of the just target. The fifth (tolerance $\sim 10\text{¢}$ at A3, $\tau = 4$ Hz) is well-approximated by any EDO whose fifth is within 10 cents of 702 cents: this includes 12-TET (2¢ error), 19-TET (7¢), 31-TET (5¢), and 53-TET (0.07¢), but excludes 5-TET (720¢, error 18¢) and 7-TET (686¢, error 16¢). The major third (tolerance $\sim 6\text{¢}$ at A3) is well-approximated only by 19-TET (1.5¢ error), 31-TET (0.8¢), and 53-TET (1.4¢); 12-TET (14¢) fails this test at most registers. This arithmetically explains the longstanding preference of renaissance-era theorists and contemporary microtonal composers for 19-TET and 31-TET: these systems preserve 5-limit harmony within the acoustic tolerance budget, whereas 12-TET can only do so for the fifth and fourth. The framework thus connects the arithmetic complexity measure $f(p/q)$ to a principled criterion for evaluating tuning systems.

The minor seventh and competing tunings

The minor seventh provides a striking special case. The Ptolemaic minor seventh 9/5 (996 cents) and the Pythagorean minor seventh 16/9 (995 cents) are only 1 cent apart, yet have distinct arithmetic characters. Gradus gives both the same value ($G = 9$), and is therefore unable to distinguish these historically competing tunings. The formula f does distinguish them: $f(9/5) = 13$ vs $f(16/9) = 20$, favouring the Ptolemaic tuning. The Pythagorean version $16/9 = (4/3)^2$ arises as two stacked fourths and is the minor seventh generated by the Pythagorean system, in which all intervals derive from the perfect fifth. Its higher f value reflects its larger numerator: the upper note sits at the 16th partial, far higher in the harmonic series than the 9th partial of the Ptolemaic version. Gradus, being symmetric in p and q , cannot see this distinction ($\Omega^*(16) + \Omega^*(9) = \Omega^*(9) + \Omega^*(16)$); the asymmetry of f is what exposes it.

10 A Partial-Beating Model of Perceptual Tolerance

The preceding section treats tolerance as a simple frequency deviation in cents. A more physically grounded model connects tolerance directly to the arithmetic structure of the interval and to the same quantity p that appears in $f(p/q)$.

The model

For a just interval p/q , the m -th coincident partial occurs at frequency $mp \cdot f_0$ (the (mp) -th harmonic of the lower note, with amplitude $\propto 1/(mp)$). If the actual interval deviates by Δ cents, the beat rate at the m -th coincidence is:

$$\beta_m \approx mp \cdot f_0 \cdot \frac{\ln 2}{1200} \cdot |\Delta| \quad \text{Hz}$$

The total roughness contribution, weighted by partial amplitude $\propto 1/(mp)$, is:

$$R \propto \sum_{m=1}^{\infty} \frac{1}{mp} \cdot \beta_m = \frac{f_0 \ln 2}{1200} \cdot |\Delta| \cdot \sum_{m=1}^M 1 = \frac{f_0 \ln 2}{1200} \cdot |\Delta| \cdot M$$

where the mp factors cancel. The sum diverges formally but is cut off in practice at the last audible partial $M \approx 10\text{--}15$, making the total roughness roughly proportional to $p \cdot f_0 \cdot |\Delta|$ — the same scaling as the first partial alone. Focusing on the first coincidence (at frequency $p \cdot f_0$) is therefore a good approximation; the higher coincidences add a logarithmic correction that does not change the $1/p$ dependence.

Setting a threshold τ Hz on the first-coincidence beat rate gives a tolerance half-width:

$$\Delta_{\text{tol}}(p/q, f_0) \approx \frac{1730 \tau}{p \cdot f_0} \text{ cents}$$

The Plomp–Levelt critical bandwidth at frequency f is approximately $\text{CB}(f) \approx 1.72 f^{0.65}$ Hz [5]; roughness is first perceptible at ~ 0.05 CB and reaches its maximum near 0.25 CB. At the first coincidence of the fifth ($p = 3$, frequency $3 \times 220 = 660$ Hz at A3), $\text{CB}(660) \approx 118$ Hz, giving roughness onset at ~ 6 Hz. This calibrates the three thresholds used below: $\tau = 4$ Hz (below roughness onset, barely perceptible), $\tau = 8$ Hz (perceptible beating, not yet rough), and $\tau = 15$ Hz (at or above roughness onset for most registers).

Two properties follow. First, **tolerance is proportional to $1/p$** : the same numerator that appears in $f(p/q)$ determines how precisely the interval must be tuned to avoid audible beating. Consonant intervals (small p , small f) are not only perceived as smooth but are also forgiving of mistuning. Second, **tolerance depends on register**: at higher pitches the tolerance in cents narrows because the beating partials are at higher absolute frequencies.

Equal temperament in the model

Figure 2 shows the tolerance half-width for each standard interval at a reference register of $f_0 = 220$ Hz (A3), for three threshold values $\tau \in \{4, 8, 15\}$ Hz. The 12-TET deviation from the just ratio is shown as a red horizontal mark.

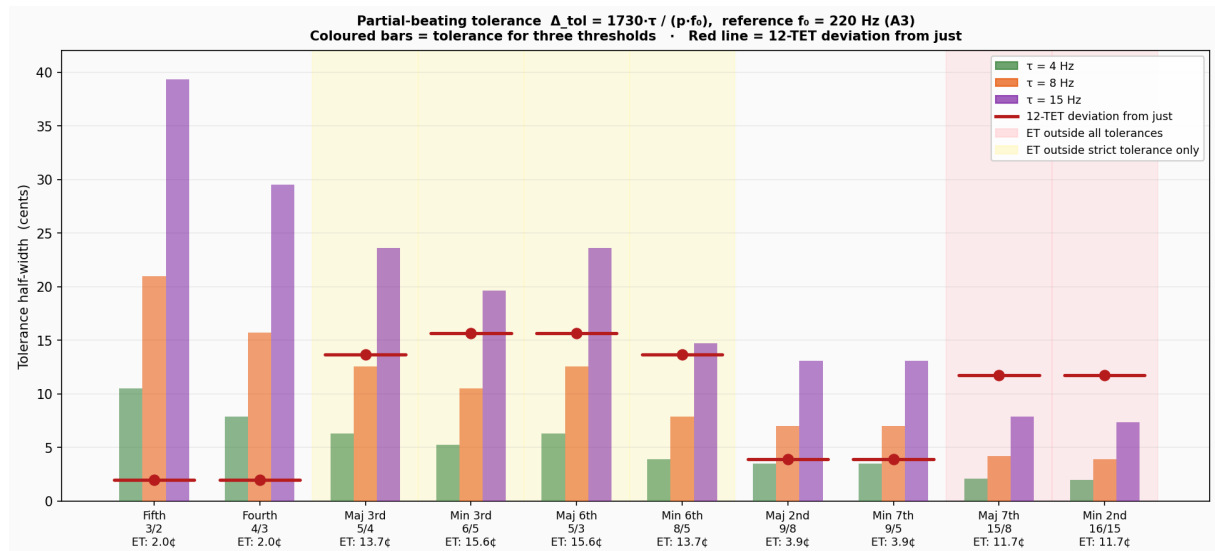


Figure 2: Partial-beating tolerance half-width vs 12-TET deviation for each just interval, at $f_0 = 220$ Hz (A3). Coloured bars show tolerance for three threshold values; the red mark is the 12-TET deviation. Yellow shading: ET exceeds the strict (4 Hz) but not the generous (15 Hz) threshold. Pink shading: ET exceeds all three thresholds.

The result is unambiguous for the fifth (3/2) and fourth (4/3): their ET deviations of 2 cents are well within tolerance at all three thresholds and at all musically relevant registers. The thirds, sixths, and sevenths fall in a different category: their ET deviations of 14–16 cents substantially

exceed the strict (4 Hz) threshold and sit near or above the generous (15 Hz) threshold. The major and minor seconds occupy an intermediate position. The major seventh (15/8) and minor second (16/15) are outside all three tolerance bands: the large numerator $p \in \{15, 16\}$ means even a modest deviation produces rapid beating.

Figure 3 plots the tolerance curves directly as functions of p , showing the $1/p$ hyperbolic structure. The interval points are coloured green (ET within all tolerances), orange (ET marginal), or red (ET outside all tolerances). The distribution confirms the pattern: the two green intervals — the fifth and fourth — are exactly those Western harmony has used as structural pillars throughout all periods of common practice, while the orange and red intervals are precisely those that show the greatest variation across tuning systems and historical periods.

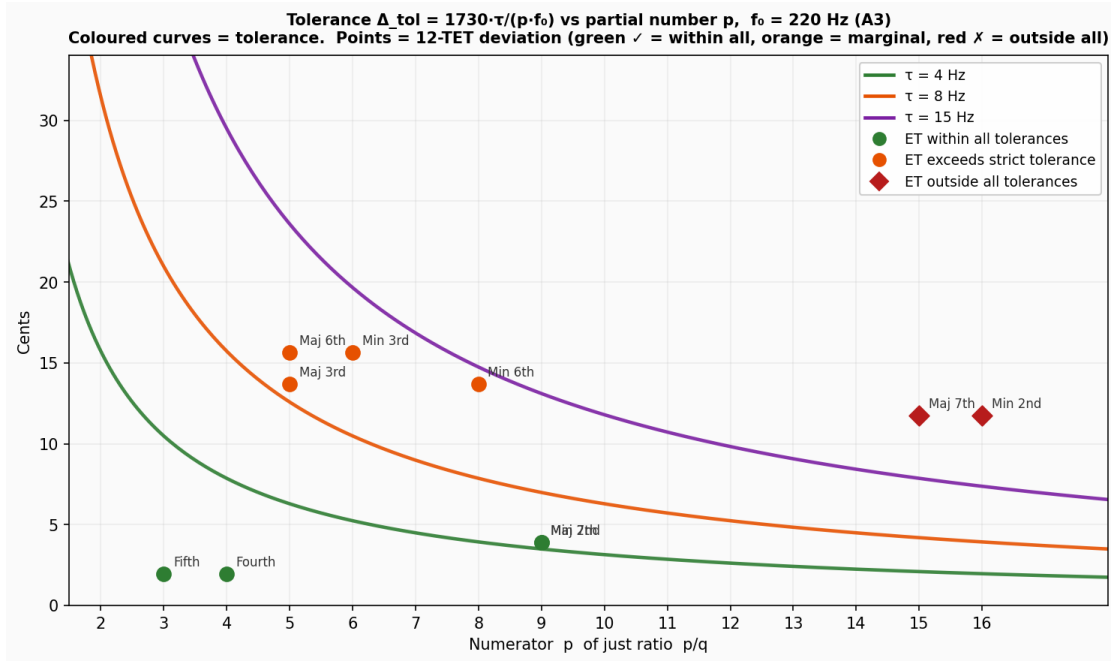


Figure 3: Tolerance half-width $\Delta_{tol} = 1730\tau/(p \cdot f_0)$ as a function of the partial number p at $f_0 = 220$ Hz. Coloured curves show three threshold values. Interval points show their 12-TET deviations: green = ET within all tolerances, orange = ET exceeds strict tolerance, red = ET outside all tolerances.

Register dependence

Figure 4 shows how tolerance varies with register for four intervals, with vertical dotted lines marking the fundamental frequency at which the ET deviation crosses each threshold. For the fifth (top left), the crossing frequency exceeds 1000 Hz for the moderate threshold — in practice the ET fifth is acoustically tolerable throughout the entire playing range of most instruments. For the major third (top right), the 15 Hz crossing occurs at around 380 Hz (roughly G4), meaning that in the upper half of the piano keyboard the ET major third produces beating comfortably above 15 Hz at the first coincident partial. For the minor third the situation is similar. The model therefore predicts that ET thirds should sound noticeably rougher in the upper register than the lower — a prediction consistent with the common practice of voicing thirds in the middle-to-lower register in orchestral writing and avoiding them in high woodwind chords.

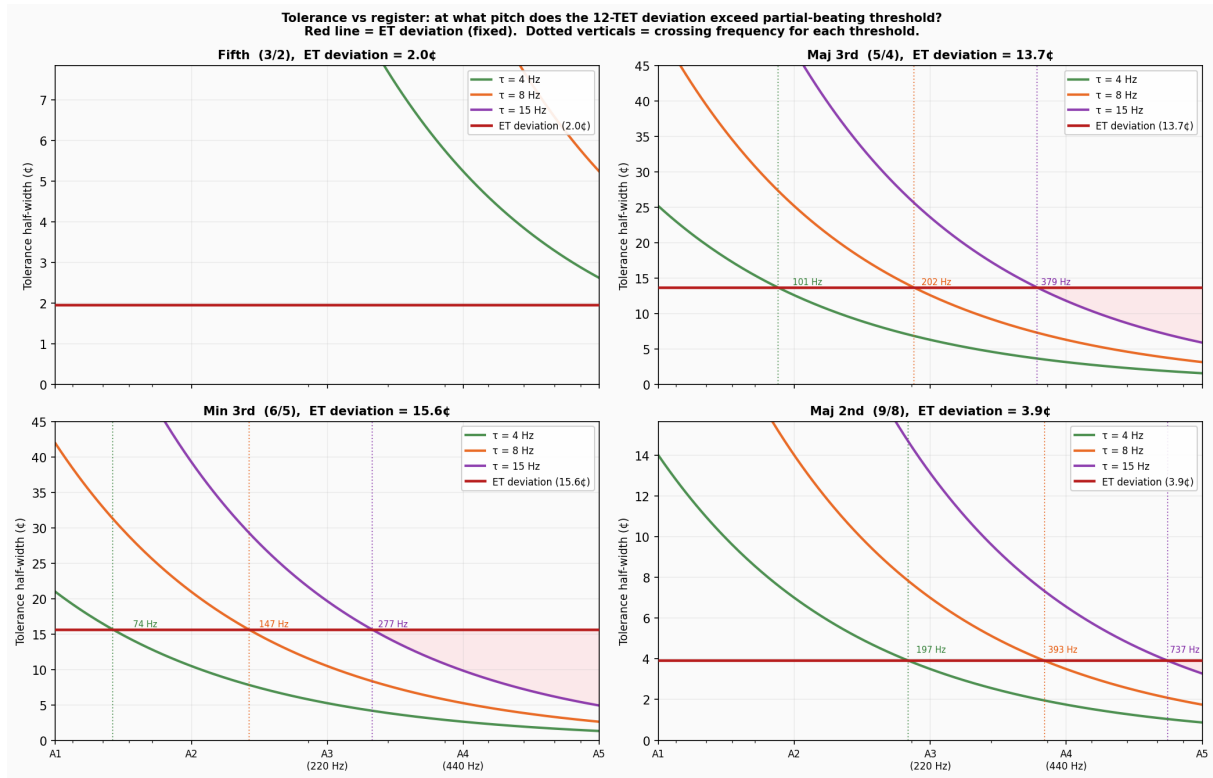


Figure 4: Tolerance half-width as a function of register (fundamental frequency, log scale) for four intervals. The red horizontal line is the fixed ET deviation; the coloured curves are the three threshold tolerances. Dotted verticals mark the register at which ET deviation crosses each threshold. Pink shading indicates the register range where ET exceeds the generous (15 Hz) threshold.

The $1/p$ duality

The model reveals a duality between the two uses of p in this framework. In the dissonance formula $f(p/q) = p + \Omega^*(q)$, the term p is the *cost* of the interval: the effort required to identify the upper note as the p -th partial of an implied fundamental. In the tolerance model, p appears in the denominator of Δ_{tol} : it is the *precision required* to keep the partial beating within an acceptable rate. These are not two separate uses of p but two aspects of the same physical fact — that p is the partial number of the upper note in the harmonic series, and that both the cognitive effort of recognition and the acoustic sensitivity to mistuning scale with partial height.

The implication for equal temperament is that Western musical culture has, over several centuries, adopted a tuning system in which the acoustically robust intervals (fifth and fourth, small p) are preserved almost exactly, while the acoustically fragile intervals (thirds and sixths, moderate p) are mistuned to the edge of — and in many registers slightly beyond — their acoustic tolerance. Whether this represents a deliberate optimisation, cultural habituation, or a combination of the two is a question that the arithmetic model alone cannot resolve.

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